



9481 - Characterizing an ongoing outburst of a rare centaur C/2023 RS61 with joint JWST and HST observations

Cycle: 4, Proposal Category: DD

INVESTIGATORS

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OBSERVATIONS

<i>Folder</i>	<i>Observation</i>	<i>Label</i>	<i>Observing Template</i>	<i>Science Target</i>
C/2023 RS61				
	1	C/2023 RS61	NIRSpec IFU Spectroscopy	(1) C2023RS61
	2	C/2023 RS61 Background	NIRSpec IFU Spectroscopy	(2) C2023RS61-OFFSET

ABSTRACT

Centaurs are small bodies scattered from the trans-Neptunian region, carrying invaluable information on the composition of trans-Neptunian objects. About 10% of centaurs show cometary activity, poorly understood and often unpredictable.

In early 2025 centaur C/2023 RS61 developed a wide tail not aligned with the anti-solar direction. Follow-up images from July 2025 show a weak, anti-solar oriented tail, while on August 10 the brightness of C/2023 RS61 suddenly increased by >4 magnitudes, indicative of a powerful outburst. Archival images reveal at least two more brightenings in the past, suggesting this object is indeed a rare distant outbursting Centaur with rapidly changing coma morphology. Moreover, it is a dynamically new centaur, meaning the nucleus has not yet been significantly thermally processed, therefore offering a unique opportunity to analyze volatiles driving a distant activity on a nearly-pristine small body, with the potential to trace back its formation location in the protosolar disc.

We propose 5.5 hours and 1 orbit respectively of joint JWST and HST observations to characterize the rare centaur C/2023 RS61 (PANSTARRS) that has undergone at least four major outbursts and displays an unusually red dust coma. The proposed program will allow to acquire high-resolution spectra of the gas species driving the current outburst complemented with the detailed inner dust coma morphology linking the gas species to their nucleographic sources.

Time-critical DD time is necessary, as C/2023 RS61 is undergoing a powerful outburst, meaning the coma brightness will likely decrease rapidly and disappear before the start of a new regular observing cycle.

OBSERVING DESCRIPTION

The observations proposed are to investigate the gas species (e.g., CO, CH₄, or CO₂) driving the activity in the recently discovered active centaur C/2023 RS61 (PANSTARRS) using the JWST NIRSpec IFU Prism mode during Cycle 4. The observations have dedicated background observations due to the extended nature of the target's gas and dust comae. The science and background observations are sequenced together and non-interruptible to ensure the most accurate measure of the background. Also, HST WFC3/UVIS imaging data are to be collected close in time ($\pm \sim 3$ days) for high-resolution inner coma morphology data for a deeper understanding of the outburst vs. persistent low-level activity.

Proposal 9481 - Targets - Characterizing an onging outburst of a rare centaur C/2023 RS61 with joint JWST and HST observations

Solar System Targets	#	Name	Level 1	Level 2	Level 3
	(1)	C2023RS61	TYPE=COMET,Q=8.000601912855959,E=0.3276831 041371935,I=19.94378143878478 ,O=347.0154081198834,W=119.3522397524865,T=30 -NOV- 2028:05:20:02,TTTimeScale=TDB,EQUINOX=J2000,E POCH=02-FEB-2025:00:00:00,EpochTimeScale=TDB		
	Comments: Extended=YES				
	(2)	C2023RS61-OFFSET	TYPE=COMET,Q=8.000601912855959,E=0.3276831 041371935,I=19.94378143878478 ,O=347.0154081198834,W=119.3522397524865,T=30 -NOV- 2028:05:20:02,TTTimeScale=TDB,EQUINOX=J2000,E POCH=02-FEB-2025:00:00:00,EpochTimeScale=TDB	TYPE=POS_ANGLE,RAD=60,ANG=90,REF=SUN	
	Comments: Extended=YES				

Proposal 9481 - Observation 1 - Characterizing an onging outburst of a rare centaur C/2023 RS61 with joint JWST and HST observati...

Observation	Proposal 9481, Observation 1: C/2023 RS61												Fri Oct 17 22:00:09 GMT 2025
	Diagnostic Status: Warning												
	Observing Template: NIRSpec IFU Spectroscopy												
	Background Observations:[C/2023 RS61 Background (Obs 2)]												
Diagnostics	(Visit 1:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.												
	(C/2023 RS61 (Obs 1)) Informational (Form): The Visit Planner and Spike may produce different schedulability results.												
Solar System Targets	#	Name	Level 1				Level 2				Level 3		
	(1)	C2023RS61	TYPE=COMET,Q=8.000601912855959,E=0.3276831041371935,I=19.94378143878478,O=347.0154081198834,W=119.3522397524865,T=30-NOV-2028:05:20:02,TTimeScale=TDB,EQUINOX=J2000,EPOCH=02-FEB-2025:00:00:00,EpochTimeScale=TDB										
	Comments: Extended=YES												
Template	TA Method						HFF Readout Mode						
	NONE						false						
Dithers	#	Dither Type		Size		Starting Point		Number of Points		Points			
	1	4-POINT-DITHER											
Spectral Elements	#	Grating/Filter	Readout Pattern	Groups/Int	Integrations/Exp	Leakcal	Dither	Autocal	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID	
	1	PRISM/CLEAR	NRSIRS2RAPID	80	1	false	true	NONE	4	4	4726.8	258870	
Special Requirements	Sequence Observations 1, 2, Non-interruptible												
	DEFAULT WINDOW: ANGULAR RATE C2023RS61 FROM JWST LESS THAN 0.075												

Proposal 9481 - Observation 2 - Characterizing an onging outburst of a rare centaur C/2023 RS61 with joint JWST and HST observati...

Observation	Proposal 9481, Observation 2: C/2023 RS61 Background											Fri Oct 17 22:00:09 GMT 2025
	Diagnostic Status: Warning											
	Observing Template: NIRSpec IFU Spectroscopy											
	Background Observation For: [C/2023 RS61 (Obs 1)]											
Diagnostics	(Visit 2:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.											
	(C/2023 RS61 Background (Obs 2)) Informational (Form): The Visit Planner and Spike may produce different schedulability results.											
Solar System Targets	#	Name	Level 1				Level 2				Level 3	
	(2)	C2023RS61-OFFSET	TYPE=COMET,Q=8.000601912855959,E=0.3276831 041371935,I=19.94378143878478 ,O=347.0154081198834,W=119.3522397524865,T=30 -NOV- 2028:05:20:02,TTimeScale=TDB,EQUINOX=J2000,EPOCH=02-FEB-2025:00:00:00,EpochTimeScale=TDB				TYPE=POS_ANGLE,RAD=60,ANG=90,REF=SUN					
	Comments: Extended=YES											
Template	TA Method						HFF Readout Mode					
	NONE						false					
Dithers	#	Dither Type		Size		Starting Point		Number of Points		Points		
	1	4-POINT-DITHER										
Spectral Elements	#	Grating/Filter	Readout Pattern	Groups/Int	Integrations/Exp	Leakcal	Dither	Autocal	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID
	1	PRISM/CLEAR	NRSIRS2RAPID	80	1	false	true	NONE	4	4	4726.8	258870
Special Requirements	Sequence Observations 1, 2, Non-interruptible											
	DEFAULT WINDOW: ANGULAR RATE C2023RS61-OFFSET FROM JWST LESS THAN 0.075											